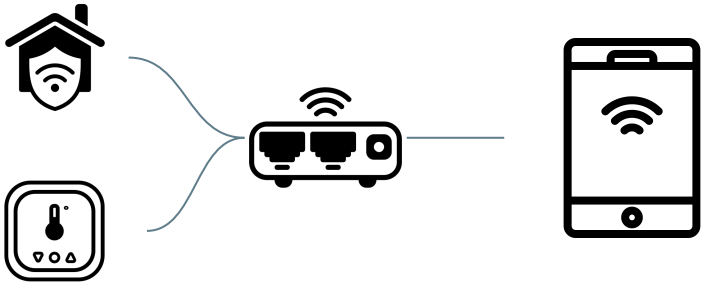
# Updating IoT System

We acknowledge funding provided by Microelectronic Commons and the support of OUSW(R&E).

# What to update?



## What to update?

- IoT device:
- Controller (App):
- Cloud:

## What to update?

- IoT device: Firmware, configuration files (e.g., certification revocation list), etc.
- Controller (App): App, permissions, configuration, etc.
- Cloud: Main application, libraries, operating system, etc.

## Definitions

- Full update: Complete new software package.
- Delta update: Difference between old and new package.
- Update installation: Applying the downloaded update to the device.
- Activation: Start execution of the newly installed software.
- Roll back: Returning to previous working version after update failure.

## Definitions

Update Latency: Time from update is available and the device runs successfully with the update.

Why this latency?

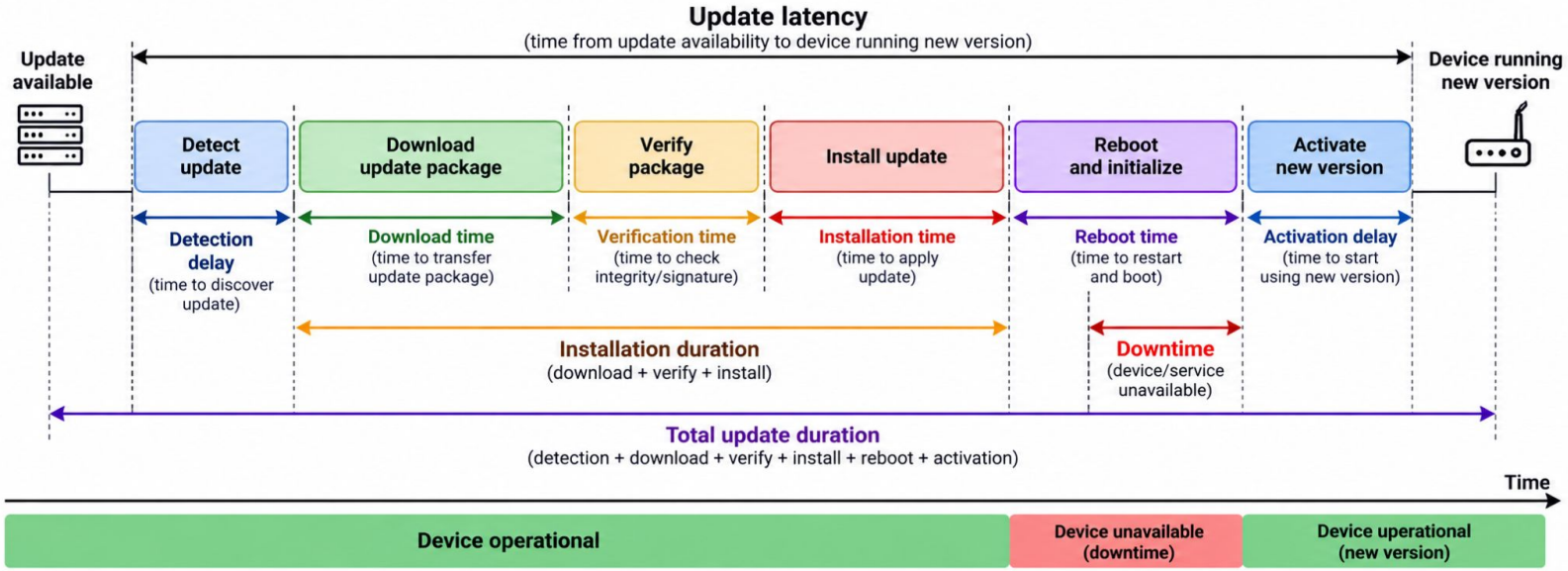
## Definitions

Update Latency: Time from update is available and the device runs successfully with the update.

Detection Delay, Download Time, Installation Time, Activation Delay, Reboot Time

# Definitions

Update Latency: Time from update is available and the device runs successfully with the update.



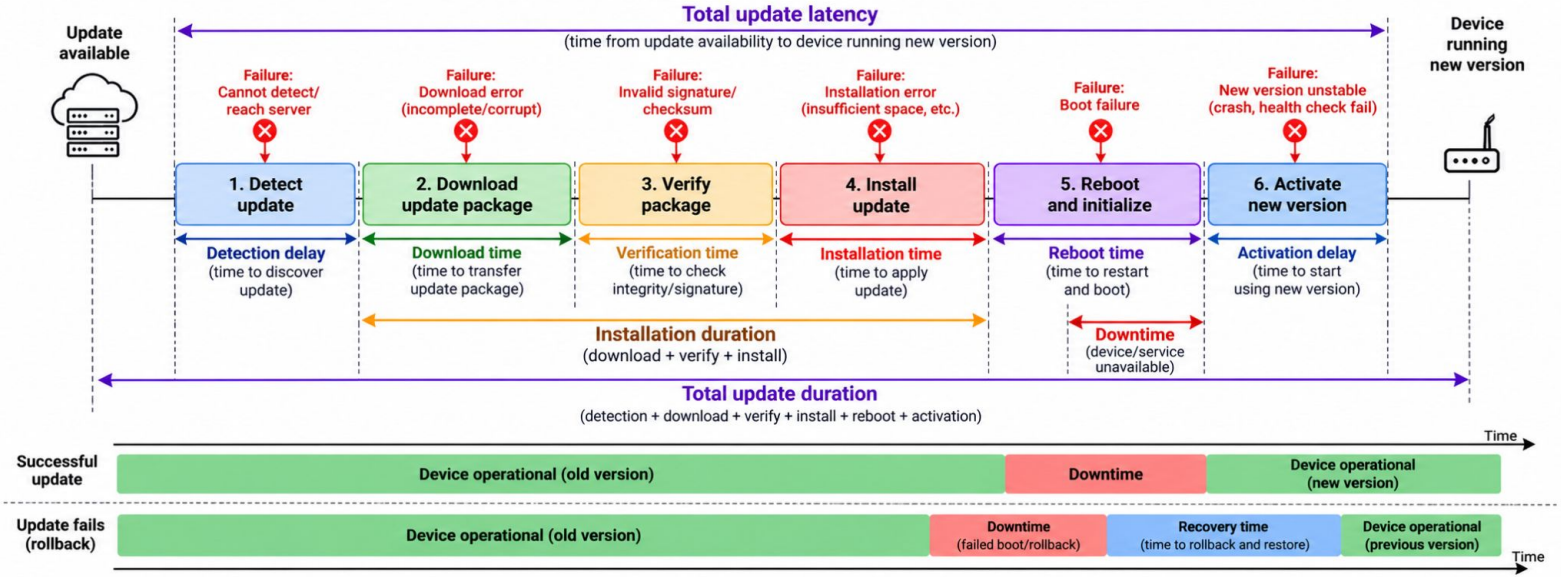
## Definitions

Update Latency: Time from update is available and the device runs successfully with the update.

Recovery Time: Time between update failure and restoring to previous known version.

# Definitions

Rollback Time: Time between update failure and restoring to previous known version.



## Software Update 101

- Check for updates
- Download the update and verify its integrity
- Install the update
- Activate the update

## Software Update 101

- Check for updates (pull, poll, or push): from where?
- Download the update and verify its integrity: to where?
- Install the update: How?
- Activate the update (mandatory, on-demand, or upon restart)

## Software Update Example: apt

- Check for updates? Pull
  - /etc/apt/sources.list
- When to update?
  - Installed version lower than available version
    - /var/lib/dpkg/status
- Install the update:
  - Download the package to /var/cache/apt/archives
- Install the .deb